

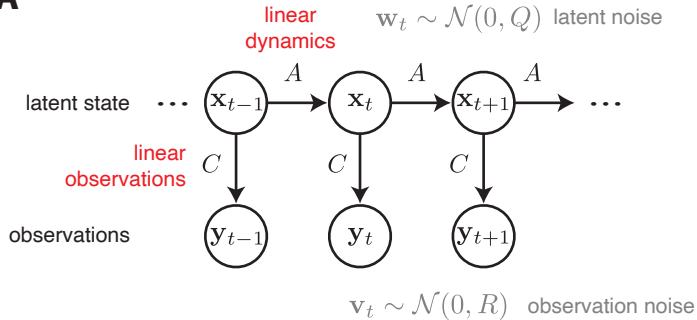
A**B**

Diagram B illustrates the matrix representation of the state-space model. The dynamics equation is given by:

$$\begin{bmatrix} \mathbf{x}_t \end{bmatrix} = \begin{bmatrix} A \end{bmatrix} \begin{bmatrix} \mathbf{x}_{t-1} \end{bmatrix} + \begin{bmatrix} \mathbf{w}_t \end{bmatrix}$$

The observation equation is given by:

$$\begin{bmatrix} \mathbf{y}_t \end{bmatrix} = \begin{bmatrix} C \end{bmatrix} \begin{bmatrix} \mathbf{x}_t \end{bmatrix} + \begin{bmatrix} \mathbf{v}_t \end{bmatrix}$$